

4.4 Waves

This section provides knowledge and understanding of wave properties, electromagnetic waves, superposition and stationary waves. The wavelength of visible light is too small to be measured directly using a ruler. However, superposition experiments can be done in the laboratory to determine wavelength of visible light using a laser and a double slit.

There are opportunities to discuss how the double-slit experiment demonstrated the wave-like behaviour of light (HSW7).

4.4.1 Wave motion

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) progressive waves; longitudinal and transverse waves	
(b) (i) displacement, amplitude, wavelength, period, phase difference, frequency and speed of a wave (ii) techniques and procedures used to use an oscilloscope to determine frequency	PAG5
(c) the equation $f = \frac{1}{T}$	
(d) the wave equation $v = f\lambda$	
(e) graphical representations of transverse and longitudinal waves	
(f) (i) reflection, refraction, polarisation and diffraction of all waves	Learners will be expected to know that diffraction effects become significant when the wavelength is comparable to the gap width.

(ii) techniques and procedures used to demonstrate wave effects using a ripple tank

- (f) (iii) techniques and procedures used to observe polarising effects using microwaves and light

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- (g) intensity of a progressive wave; $I = \frac{P}{A}$;
intensity $\propto$ (amplitude)².